

Token-Accounting Integrity: Measuring Client-Side Under-Payment in LLM Metering Architectures

(anonymized for review)

Abstract

Usage-based billing is the default commercial model for LLM inference, yet the accounting pipeline that converts delivered tokens into money is itself a security boundary. Prior work on LLM billing studies a dishonest *provider* over-charging an honest client, a third party inflating a *victim's* bill, and a dishonest *intermediary* forging provenance. We study the remaining direction: an honest provider and a dishonest *client* that seeks inference value while under-paying. We argue this is not a list of attacks but one accounting boundary with three placements — **state synchronization**, **economic commitment timing**, and **usage authority** — and we model each as a transition of a request-lifecycle state machine with a single integrity property, $V(r) \leq N(r)$. We do not claim new attack primitives; the enabling mechanisms are individually known. On a vulnerable-by-construction testbed with integrity gates that independently re-derive every reported number, we measure all three across architectures, concurrency, abort position, price tier, tokenizer, and workload. Leakage from commitment-timing and usage-authority defects is *invariant* to concurrency (identical to four decimal places from 1 to 100 concurrent requests), whereas the synchronization defect is *created* by it (slope 0.099 per additional concurrent request, $R^2 = 1.0000$). An ablation shows different primitives close different properties: abort-safe finalization closes accounting integrity while reservation independently provides solvency, and computing a server-side recount is insufficient unless it is also the billing basis — an architecture that computes a correct recount and bills the client's number still leaks 58.3% of the delivered value. A local real-model experiment bounds the strongest defense's cost at 0.014–0.019% of end-to-end inference time, correcting a 0.05–0.76 $\times$ throughput figure measured against a zero-cost mock generator. All results regenerate from raw data in a 56-minute clean run.

1 Introduction

Because the billed unit — the token — is produced by opaque, streamed, long-running inference, the pipeline that authorizes, meters, and commits a charge is security-relevant. Recent work studies dishonest *providers* inflating that accounting [1, 2, 3], economic attacks that drain a victim's budget [4, 5], and dishonest *intermediaries* that misroute or forge provenance [10, 11]. All three assume the client is honest, impersonated, or irrelevant. We study the remaining direction (Table 1): a legitimate but dishonest client that under-pays.

We are deliberately adversarial about our own contribution. Each enabling mechanism is already known in isolation: a deployed LLM gateway refunds pre-consumed quota when a final usage record is missing after a client disconnect [6]; “never trust client input in a security decision” is CWE-807, and a metered-token proxy already mitigates an encoding-induced “silent zero-debit” [7]; concurrent quota-decrement races are a classic web-race primitive with a recent CVE [8, 9]; and practitioner guidance already warns that retries and disconnects must not cause token leakage [12]. Accordingly,

Table 1: Adversary directions in LLM inference accounting. Prior work covers the first three; this paper studies the fourth.

Adversary	Goal	Representative work	Studied here
Provider	over-charge an honest client	CoIn [1], Invisible Tokens [2], Token Inflation [3]	no
Third party	inflate a <i>victim</i> ’s bill	Denial-of-Wallet [4], LLMjacking [5]	no
Intermediary	misroute / forge provenance	AEX [10], gateway-path provenance [11]	no
Client	under-pay for received inference	—	yes

M1 and M2 are systematization and measurement contributions, and B0 is a known baseline; we make no priority claim.

Contributions.

1. **An accounting model** (§4) connecting inference execution, value delivery, authorization, reservation, debit, reconciliation and refund in a request-lifecycle state machine, with a formal integrity property that admits legitimate refunds.
2. **An architecture taxonomy** of three experimentally distinguishable dimensions — synchronization (B0, known baseline), commitment timing (M1), and usage authority (M2) — with the known/systematized/defensive boundaries stated explicitly.
3. **A reproducible measurement study:** a testbed and methodology quantifying attack success, leakage, leakage efficiency, detectability, invariant violations, latency and throughput, with integrity gates proven to fail closed and an independent re-implementation of the strongest defense.
4. **An integrity/performance trade-off**, including an ablation identifying which primitive closes which property, and recount cost measured at three levels (standalone tokenizer, gateway, and a local real-model experiment).

2 Threat Model and Accounting Invariant

Honest provider; dishonest client holding valid credentials. The client controls exactly six things, each exercised by our harness: request payloads including any client-declared usage fields; permitted headers; request concurrency; the connection lifecycle (it may abort at any moment); retries; and observation of its own responses and balance. It *cannot* compromise TLS, access the database or Redis, read the gateway filesystem, compromise the inference backend, forge authenticated upstream metadata, or alter server code. This is a concrete application-layer adversary, not a symbolic cryptographic one. Every leak we report is therefore attributable to the provider’s own accounting design rather than to a compromise.

For a request r , let $V(r)$ be the value actually delivered, $D(r)$ the committed debits, $F(r)$ refunds, and $N(r) = D(r) - F(r)$ the net committed debit. Under-payment and the integrity property are

$$\text{Leak}(r) = V(r) - N(r), \quad \text{Integrity}(r) : V(r) \leq N(r).$$

The property is an inequality on the *net* debit precisely so that legitimate reconciliation is admitted: a client that consumes less than a reservation must be refunded the difference. A refund is legitimate only when it releases the unconsumed portion, $F(r) \leq \text{Res}(r) - V(r)$; the vulnerable “refund the whole reservation on abort” architecture violates exactly this side condition.

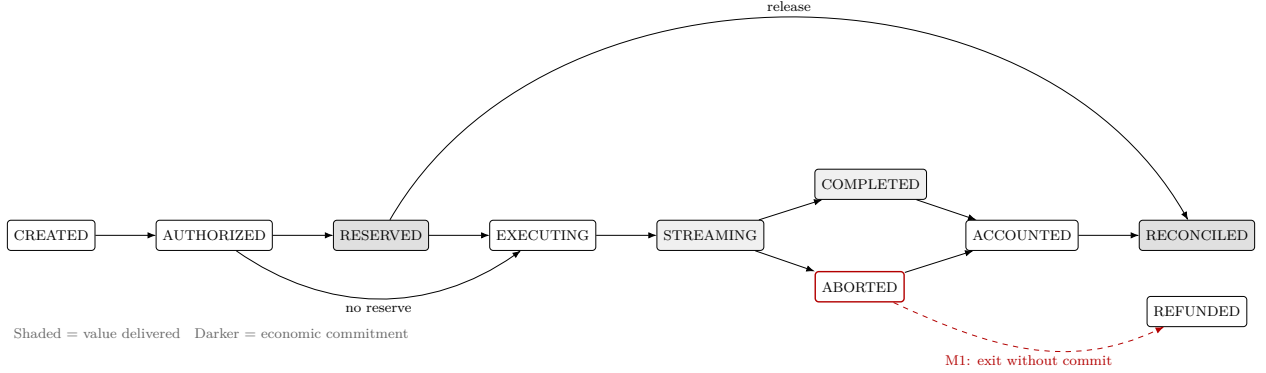


Figure 1: Request lifecycle. Value exposure begins at the first delivered token (STREAMING); economic commitment is final only at RECONCILED. The interval between them is the exposure window that M1 exploits. B0 corrupts the AUTHORIZED→RESERVED transition (non-atomic); M2 corrupts the *data* used at ACCOUNTED.

3 Background and Related Work

LLM APIs meter multi-category token vectors (input, output, cached, reasoning), stream output incrementally, and settle charges asynchronously. Provider-side auditing work establishes that hidden-token accounting can be inflated and is hard to verify [1, 2, 3]; Denial-of-Wallet and LLM-jacking establish victim-side economic exposure [4, 5]; attestation work establishes intermediary-side provenance [10, 11]. Generic quota races are well understood [8, 9]. What we found no prior study of, *in our targeted corpus*, is an evaluation of client-side under-payment under an honest-provider threat model across representative metering architectures with a common measurement and an ablation of the enforcement primitives. This is a bounded negative from a targeted search; it does not establish non-existence.

4 Accounting State Model

We model a metered inference request as a state machine so that each mechanism can be localized to a specific transition, establishing that the taxonomy is *architectural* rather than a list of unrelated implementation bugs.

4.1 Formal model

For a request r let $U_{\text{true}}(r)$ be the usage actually produced, $U_{\text{bill}}(r)$ the usage billed, and $P(\cdot)$ the (exact-decimal) pricing function. Write $C(r) = P(U_{\text{true}}(r))$ for the true cost, $V(r)$ for the value actually delivered to the client, $\text{Res}(r)$ for any reservation, $D(r)$ for committed debits, $F(r)$ for refunds, and

$$N(r) = D(r) - F(r)$$

for the *net committed debit*. Under-payment and the integrity property are

$$\text{Leak}(r) = V(r) - N(r), \quad \text{Integrity}(r) : V(r) \leq N(r).$$

Table 2: Structural distinction between the three mechanisms.

Mechanism	Defective element	Needs concurrency?	Closed by
B0	transition atomicity	yes (≥ 2)	atomic compare-and-decrement
M1	transition graph (abort path)	no	abort-inclusive finalization
M2	data authority at ACCOUNTED	no	server-authoritative recount

Because legitimate reconciliation exists, integrity is an inequality on the *net* debit; a refund is legitimate only when it releases the unconsumed portion,

$$F(r) \leq \text{Res}(r) - V(r).$$

At trial level the harness independently verifies conservation,

$$\sum_{r \in T} N(r) = B_{\text{initial}} - B_{\text{final}},$$

and fails the build on any divergence.

4.2 Mechanism localization

B0 (synchronization). If two concurrent requests read the same authorization state B and both commit a decrement derived from it, one decrement is lost: $\sum_i N(r_i) < \sum_i V(r_i)$. This requires ≥ 2 concurrent transactions, which is why B0 leaks nothing at concurrency 1.

M1 (commitment timing). If $V(r) > 0$ in STREAMING but the exit path taken (ABORTED) does not reach RECONCILED with $N(r) \geq V(r)$, then $\text{Leak}(r) = V(r) > 0$. The defect is in the transition graph, not in a shared resource, hence it is concurrency-invariant.

M2 (usage authority). If the billing basis is client-controlled, $U_{\text{bill}} = T(U_{\text{true}})$ for an attacker transformation T , then $\text{Leak}(r) = P(U_{\text{true}}) - P(T(U_{\text{true}})) > 0$ whenever $P \circ T < P$. Authority is a per-request data-provenance property, hence also concurrency-invariant.

This predicts the empirical result: B0 leakage is *created* by concurrency, while M1/M2 per-request leakage is invariant to it.

5 Taxonomy and Testbed

The three dimensions are experimentally distinguishable, not merely conceptual: **B0** (synchronization) is a non-atomic authorize-then-decrement, is *concurrency-created*, and is labelled a **known baseline** used to validate the measurement rig and to provide a generic reference point; **M1** (commitment timing) is lifecycle-sensitive; **M2** (usage authority) is authority- and pricing-sensitive. Applying the LLM-specificity test explicitly — *would this attack work unchanged against a non-LLM API?* — B0 answers *yes* (hence baseline), while M1 depends on streamed, long-running delivery and M2 on multi-category token pricing, so both answer *no*.

The testbed is a FastAPI gateway over PostgreSQL 16 (READ COMMITTED) and Redis 7, with a deterministic mock generator for reproducibility and *real* tokenizers on the recount path, run under Docker Compose on one commodity host (16 logical CPUs, 16 GB). Money is exact decimal. Five M1 architectures and six M2 architectures share common interfaces so that experiments compare *semantics* rather than unrelated code.

6 Experimental Methodology

Category-wise pricing over three tiers; leakage *efficiency* (fraction of true value evaded) is the price-invariant headline metric and absolute dollars are labelled scenario values. Sweeps: B0 concurrency $\{1, 2, 5, 10, 20, 30, 50\} \times 30$ repetitions; M1 architecture $\times$ abort position $\times$ concurrency $\{1, 5, 20, 50, 100\}$; M2 architecture $\times$ manipulation $\times$ concurrency. Request volume is held *constant* across concurrency levels so that volume cannot masquerade as vulnerability.

Integrity gates. Every summarizer independently recomputes each record’s leak from primitives, checks invariant-flag consistency, and verifies ledger conservation $\sum_r N(r) = B_{\text{init}} - B_{\text{final}}$ without trusting the application’s own bookkeeping; a violation fails the build. We validate the gates adversarially: corrupting a leak value, a debit, a served flag, a balance, an invariant flag, or deleting a ledger row must each be detected (12/12), alongside metamorphic properties (10/10). The metamorphic suite also caught one of our own unjustified assumptions — we had asserted leakage efficiency was price-tier invariant, but that holds only under *uniform* price scaling, not under a structural change to the price ratios — which we corrected rather than suppressed.

Independent verification. The strongest M2 defense is re-implemented in a module sharing no code path with the gateway; verdicts agree on all cross-validated cells (42 M2, 16 M1, 2 B0). The independent checker initially reported three disagreements, which were *definitional*, not defects: it aggregated under-payment only, while the gateway aggregates the signed billing delta. Under flat-total pricing an honest client is systematically *overcharged*, so the two aggregates legitimately differ. We now report both, and use under-payment for the security property.

Statistics. All statistics are produced by a standalone script (scipy/statsmodels) into `stats_output.txt`, which the paper cites. Many conditions here are deterministic by construction and have exactly zero variance; for those we report the constant and complete separation rather than a p-value, and reserve nonparametric tests and bootstrap intervals for quantities that genuinely vary.

7 Results

7.1 B0 — synchronization (known baseline)

The non-atomic check-then-decrement leaks only when concurrency ≥ 2 : mean \$-leak is 0.000 at $c=1$, then 0.099, 2.871 and 4.851 at $c=2, 30$ and 50 , growing exactly linearly (slope 0.099 per additional concurrent request, $R^2 = 1.0000$, $p = 1.9 \times 10^{-50}$). The hardened atomic compare-and-decrement leaks \$0 at every level (Table 3, Fig. 5). B0 is not a contribution; it validates the rig and provides the contrast used below.

7.2 M1 — commitment timing

Leakage tracks the value delivered before the missing commit, rising monotonically with abort position and collapsing to zero at completion when the debit finally fires: per request 0.0045 at ~ 1 token delivered, 0.0285 at 25%, 0.0540 at 50%, 0.0960 at 90%, and 0.0000 at 100% (Fig. 2, Table 4). Request-ASR is 1.00 at every abort position before completion. The two vulnerable architectures — “debit only after completion” and “refund the whole reservation on abort” — are economically identical.

M1 comprises two orthogonal security properties. *Property A (solvency):* served value must not exceed the affordable balance. *Property B (accounting finality):* delivered value must not exceed the net committed debit. The ablation separates them (Table 6). With a budget for two requests and six issued, `no_reserve_settle` preserves accounting integrity (leak 0.0000 at every abort position) yet serves all six and drives the balance to -0.32 ; conversely `reserve_refund_on_abort` holds a reservation but refunds it away, failing both. **Reservation and abort-safe finalization are orthogonal controls; neither substitutes for the other.**

Concurrency is not the mechanism. Holding request volume constant, leak per request is 0.0528 at $c = 1, 5, 20, 50$ and 100 — identical to four decimal places — and request-ASR is exactly 1.000 (vulnerable) or 0.000 (safe) at every level (Table 5, Fig. 3). The honest-client control arm leaks zero everywhere.

7.3 M2 — usage authority

Writing U for the true usage vector, P for the pricing function and T for the attacker’s transformation of the declared usage, a manipulation pays off exactly when $P(T(U)) < P(U)$ — that is, when the pricing basis *reads the field T changes*. Table 9 derives this per manipulation: a total/subtotal mismatch is inert under category pricing (which never reads the declared total) and is the strongest attack under flat-total pricing (0.741), while reclassifying output into a discounted cached category is the reverse. The exploitable set is a property of the billing function, not of attacker ingenuity. Every server-authoritative architecture drives leakage efficiency to 0.000 across all manipulations, and M2 is likewise concurrency-invariant (Table 8).

Usage computation is not usage authority. The decisive case is `client_logged`: it *computes* a correct server-side recount, logs it, and still leaks at efficiency **0.583**, because billing continues to use the client’s number (Table 12). A correct recount that is not authoritative for billing buys detection, not prevention.

Economic sensitivity. Leakage efficiency is exactly invariant under uniform price scaling ($\times 10$) and shifts by at most 0.012 under a price-*structure* change, so the qualitative conclusions are price-independent (Table 10). A side effect of the same asymmetry is that flat-total pricing systematically *overcharges* honest clients (efficiency -0.058).

Concurrency does not amplify these failures. Increased concurrency is not itself a sufficient amplifier of client-side metering evasion. In the tested architecture, B0 exhibited a concurrency-created race, whereas M1 and M2 remained primarily architectural accounting failures. Reporting this negative result matters: it means an operator cannot infer safety from low traffic, and cannot detect M1/M2 by watching for load spikes.

7.4 Lifecycle failure modes

Beyond the scripted attack we inject nine lifecycle failures (disconnect at $t=0$ and mid-stream, truncated stream, settle-window race, missing / delayed / duplicated usage metadata, and a recount-engine failure). Every invariant violation falls on a vulnerable architecture; the hardened architectures survive all of them; and a recount-engine failure is *fail-closed* — the request is rejected and nothing is served or billed (Table 13).

7.5 Detectability

Detectability is orthogonal to exploitability: a failure can be exploitable and hidden, exploitable but reconcilable, or prevented outright. We classify D0 (no evidence), D1 (post-hoc reconciliation only), D2 (logged discrepancy), D3 (prevented in real time). Under `client` billing the leak is D0 — the ledger faithfully records the client’s lie. Merely *logging* a recount moves the same leak to D1 without preventing it, and reconciling in-request reaches D3 (Table 14). This gives operators who cannot yet re-architect billing a cheap intermediate step.

8 Defense Evaluation

Each vulnerable architecture is paired with a server-side primitive: atomic compare-and-decrement (B0), reserve-then-abort-safe-finalization (M1), and server-authoritative recount (M2). Under every defense the invariant-violation rate and leakage efficiency are zero across the full attack catalogue.

What the strongest defense costs. We measure recount at three levels because they answer different questions.

1. **Standalone tokenizer** (Fig. 6): cost is linear in input length (log-log slope 0.999–1.14, $R^2 \geq 0.998$), so it budgets as $\text{tokens} \times \text{a per-token constant}$ ($\approx 0.7 \mu\text{s}/\text{token}$ for tiktoken, $\approx 2.5 \mu\text{s}/\text{token}$ for SentencePiece). At 16,384 tokens the engines span 10.1–41.8ms, a 2.9–3.6 $\times$ implementation effect; at a fixed token count, workload changes cost by up to 3.4 $\times$.
2. **Gateway with a *mock* generator**: recount is nearly free at $\leq 1,024$ tokens ($\geq 0.96\times$ throughput) but costs heavily at 16,384 tokens ($0.05\text{--}0.76\times$ at $c \in \{1, 10\}$). Tokenization time explains this (Spearman $\rho = -0.868$, $p = 1.3 \times 10^{-5}$).
3. **Local real-model experiment**: with real autoregressive generation the recount is **0.014%–0.019%** of end-to-end time across 128-, 1024- and 4096-token contexts, and measured throughput ratios scatter around 1.0 with signs that flip across workloads, i.e. within a $\pm 10\%$ noise band that is itself $\sim 700\times$ larger than the effect.

The apparent expense in (2) is an artifact of the mock generator costing *nothing*: when generation is free, tokenization is the entire request. We report all three rather than the most convenient one, and we do *not* claim recount is “essentially free”: it is a real, predictable cost that is small *relative to real inference* in the tested configuration.

9 Discussion

Three placements of one accounting boundary behave differently under load, and the difference is predicted by *which* element of the state machine is defective: transition atomicity (B0) is a shared-resource property and therefore concurrency-created, whereas the transition graph (M1) and data authority (M2) are per-request properties and therefore concurrency-invariant. The practical consequences are that (i) an operator cannot use traffic level as a proxy for exposure, (ii) integrity and solvency require different controls, and (iii) a correct recount is worthless for prevention unless it is authoritative for billing. The defenses themselves are ordinary; the contribution is knowing precisely which one is load-bearing and what it costs.

Table 3: B0 baseline: credit-decrement race (30 reps/cell).

posture	conc	served	\$-leak	trial ASR	req ASR	detect
hardened	1	1.0	0.0000	0.00	0.000	none
hardened	2	2.0	0.0000	0.00	0.000	none
hardened	5	5.0	0.0000	0.00	0.000	none
hardened	10	10.0	0.0000	0.00	0.000	none
hardened	20	10.0	0.0000	0.00	0.000	none
hardened	30	10.0	0.0000	0.00	0.000	none
hardened	50	10.0	0.0000	0.00	0.000	none
vulnerable	1	1.0	0.0000	0.00	0.000	none
vulnerable	2	2.0	0.0990	1.00	0.500	partial
vulnerable	5	5.0	0.3960	1.00	0.800	partial
vulnerable	10	10.0	0.8910	1.00	0.900	partial
vulnerable	20	20.0	1.8810	1.00	0.950	partial
vulnerable	30	30.0	2.8710	1.00	0.967	partial
vulnerable	50	50.0	4.8510	1.00	0.980	partial

10 Threats to Validity

Internal. Single uvicorn worker, so gateway recount numbers are an upper bound; shared host and Docker scheduling; local loopback adds a ~ 45 ms floor per request on the Windows/Docker Desktop configuration, which makes the gateway ratios conservative for small inputs; mock generation everywhere except the real-model experiment; leakage that depends on delivered tokens is sensitive to disconnect timing at high concurrency (the within-cell standard deviation of delivered tokens is exactly zero at $c \leq 20$ and ≈ 0.011 at $c \in \{50, 100\}$).

Gateway measurements at $c \geq 50$. We exclude these from headline claims. The repeat spread reaches 17% of the median and the no-recount baseline is itself non-monotonic, indicating client-side and event-loop saturation rather than server cost; single runs even produced ratios > 1 , which is physically impossible for added work, and motivated the switch to median-of-three repeats. Only $c = 1$ and $c = 10$ are treated as evidence.

Construct. Synthetic price tiers; a token-count abstraction of “inference value”; leakage normalized by delivered value; and a detectability scale that separates exploitability from evidence availability.

External. One local architecture family, one database, one cache, one small CPU model, no commercial provider, no multi-region or distributed gateway. The real-model experiment bounds a *ratio* in one configuration and does not validate any commercial provider’s billing behaviour.

Table 4: M1 metering-commit timing: leak vs. abort (medium tier, concurrency=1).

architecture	abort%	delivered	leak/req	req ASR	inv.viol	detect
pre_debit	0	1	0.0000	0.00	0.00	n/a
pre_debit	25	17	0.0000	0.00	0.00	n/a
pre_debit	50	34	0.0000	0.00	0.00	n/a
pre_debit	75	52	0.0000	0.00	0.00	n/a
pre_debit	90	62	0.0000	0.00	0.00	n/a
pre_debit	100	69	0.0000	0.00	0.00	n/a
reserve_reconcile	0	1	0.0000	0.00	0.00	n/a
reserve_reconcile	25	17	0.0000	0.00	0.00	n/a
reserve_reconcile	50	34	0.0000	0.00	0.00	n/a
reserve_reconcile	75	52	0.0000	0.00	0.00	n/a
reserve_reconcile	90	62	0.0000	0.00	0.00	n/a
reserve_reconcile	100	69	0.0000	0.00	0.00	n/a
post_completion	0	1	0.0045	1.00	1.00	D0
post_completion	25	17	0.0285	1.00	1.00	D0
post_completion	50	34	0.0540	1.00	1.00	D0
post_completion	75	52	0.0810	1.00	1.00	D0
post_completion	90	62	0.0960	1.00	1.00	D0
post_completion	100	69	0.0000	0.00	0.00	n/a
reserve_refund_on_abort	0	1	0.0045	1.00	1.00	D0
reserve_refund_on_abort	25	17	0.0285	1.00	1.00	D0
reserve_refund_on_abort	50	34	0.0540	1.00	1.00	D0
reserve_refund_on_abort	75	52	0.0810	1.00	1.00	D0
reserve_refund_on_abort	90	62	0.0960	1.00	1.00	D0
reserve_refund_on_abort	100	69	0.0000	0.00	0.00	n/a

Literature. A targeted search rather than a proof of absence, in a fast-moving area with terminology mismatch risk.

11 Responsible Disclosure and Ethics

This work produced no new vulnerability findings against any third-party, live, or production system; all experiments ran against the local testbed shipped with the artifact, whose harness defaults to `localhost` and contains no third-party endpoints. No credentials, real user data, or production billing APIs were used. The public artifacts we cite as prior evidence (a GitHub issue [6], a PR [7], and a CVE [9]) were public before this work and are not our findings. Every attack in the artifact is paired with a defense, and the testbed is released so operators can measure their own gateways.

12 Conclusion

Client-side under-payment in LLM metering is best modelled as one accounting boundary with three placements: synchronization, commitment timing, and usage authority. Localizing each to a

Table 5: M1: leak PER REQUEST vs concurrency (request volume held constant, attacker arm).

architecture	c=1	c=5	c=20	c=50	c=100
post_completion	0.0528	0.0528	0.0528	0.0528	0.0528
pre_debit	0.0000	0.0000	0.0000	0.0000	0.0000
reserve_reconcile	0.0000	0.0000	0.0000	0.0000	0.0000
reserve_refund_on_abort	0.0528	0.0528	0.0528	0.0528	0.0528

Table 6: M1 architecture matrix. Solvency (served value must not exceed the affordable balance) and accounting integrity (delivered value must not exceed the net committed debit) are ORTHOGONAL: reservation provides the former and abort-safe finalization the latter, and neither substitutes for the other.

architecture	Property A: solvency	Property B: accounting integrity	how the violated property fails
pre_debit	HOLDS	HOLDS	-
reserve_reconcile	HOLDS	HOLDS	-
reserve_refund_on_abort	VIOLATED	VIOLATED	served 6/6 on a 2-request budget
no_reserve_settle	VIOLATED	HOLDS	served 6/6 on a 2-request budget; balance -0.320000
post_completion	VIOLATED	VIOLATED	served 6/6 on a 2-request budget

state transition predicts — and our measurements confirm — that only the synchronization defect is concurrency-created, that integrity and solvency are closed by different primitives, and that a server-side recount must be authoritative rather than merely computed. In the tested configuration that defense costs a small, predictable fraction of real inference time.

References

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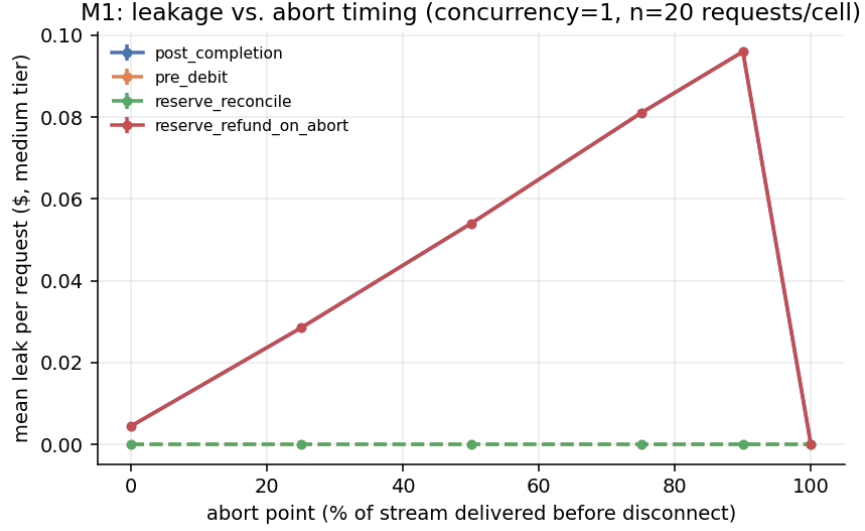


Figure 2: M1: leakage vs. abort position (concurrency 1, medium tier). Vulnerable architectures peak just before the commit and collapse to \$0 at completion; safe architectures stay at \$0. Error bars are within-cell standard deviations.

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- [12] Industry metering guidance: gateway as enforcement point vs. meter as accounting point; retries, partial failures and client disconnects must not cause double billing or token leakage, 2026.

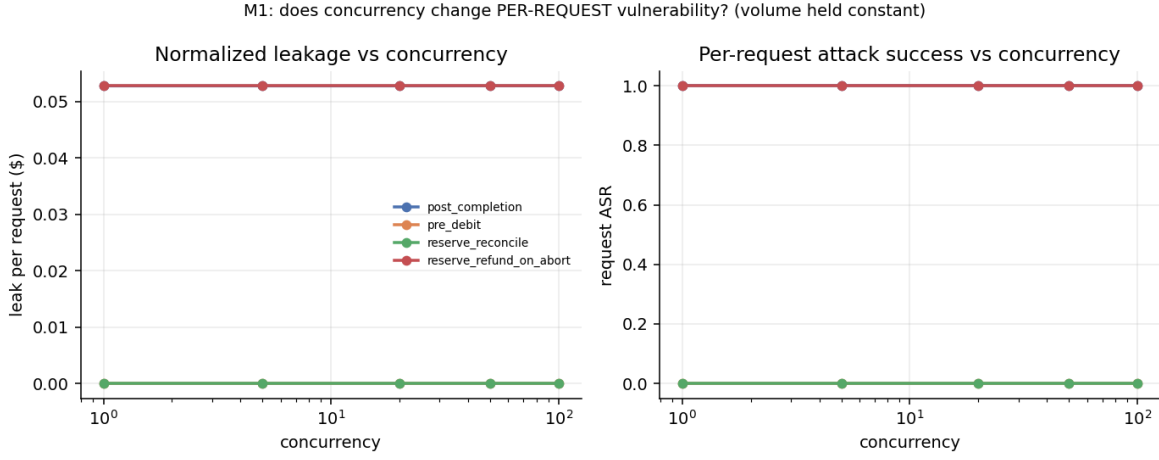


Figure 3: M1 negative result: with request volume held constant at 20 requests per cell, per-request leakage and request-ASR are invariant to concurrency from 1 to 100.

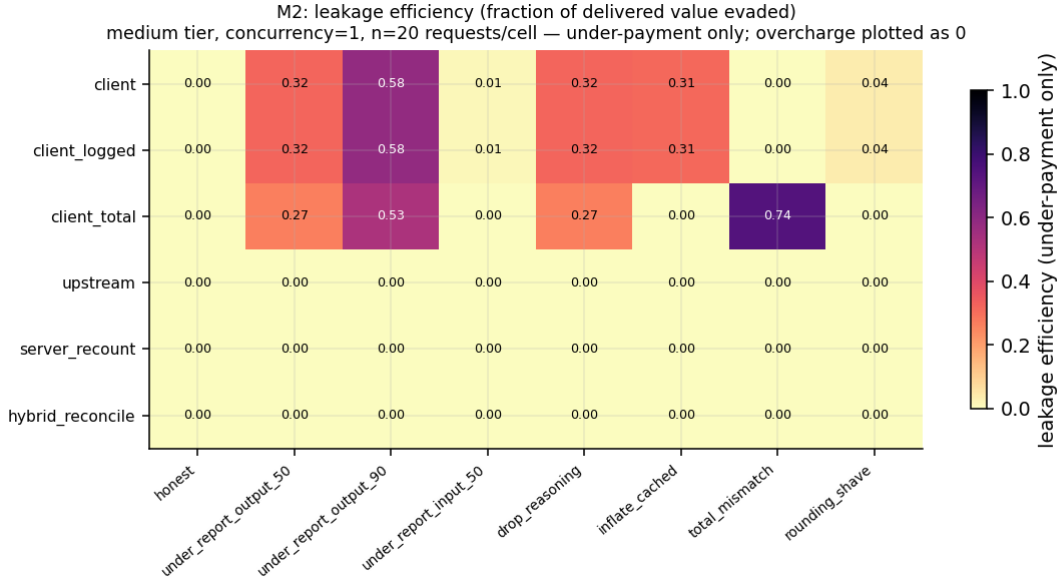


Figure 4: M2: leakage efficiency (fraction of delivered value evaded) by architecture $\times$ manipulation, medium tier, concurrency 1. Server-authoritative rows are uniformly zero; the exploitable set differs between category and flat-total pricing.

Table 7: M2 usage-record authority: leakage efficiency by manipulation (medium tier, concurrency=1).

architecture	manipulation	leak eff.	req ASR	inv.viol	detect
client	honest	0.00	0.00	0.00	n/a
client	under_report_output_50	0.32	1.00	1.00	D0
client	under_report_output_90	0.58	1.00	1.00	D0
client	under_report_input_50	0.01	1.00	1.00	D0
client	drop_reasoning	0.32	1.00	1.00	D0
client	inflate_cached	0.31	1.00	1.00	D0
client	total_mismatch	0.00	0.00	0.00	n/a
client	rounding_shave	0.04	1.00	1.00	D0
client_logged	honest	0.00	0.00	0.00	n/a
client_logged	under_report_output_50	0.32	1.00	1.00	D1
client_logged	under_report_output_90	0.58	1.00	1.00	D1
client_logged	under_report_input_50	0.01	1.00	1.00	D1
client_logged	drop_reasoning	0.32	1.00	1.00	D1
client_logged	inflate_cached	0.31	1.00	1.00	D1
client_logged	total_mismatch	0.00	0.00	0.00	n/a
client_logged	rounding_shave	0.04	1.00	1.00	D1
client_total	honest	0.00	0.00	0.00	n/a
client_total	under_report_output_50	0.27	1.00	1.00	D0
client_total	under_report_output_90	0.53	1.00	1.00	D0
client_total	under_report_input_50	0.00	0.00	0.00	n/a
client_total	drop_reasoning	0.27	1.00	1.00	D0
client_total	inflate_cached	0.00	0.00	0.00	n/a
client_total	total_mismatch	0.74	1.00	1.00	D0
client_total	rounding_shave	0.00	0.00	0.00	n/a
upstream	honest	0.00	0.00	0.00	n/a
upstream	under_report_output_50	0.00	0.00	0.00	n/a
upstream	under_report_output_90	0.00	0.00	0.00	n/a
upstream	under_report_input_50	0.00	0.00	0.00	n/a
upstream	drop_reasoning	0.00	0.00	0.00	n/a
upstream	inflate_cached	0.00	0.00	0.00	n/a
upstream	total_mismatch	0.00	0.00	0.00	n/a
upstream	rounding_shave	0.00	0.00	0.00	n/a
server_recount	honest	0.00	0.00	0.00	n/a
server_recount	under_report_output_50	0.00	0.00	0.00	n/a
server_recount	under_report_output_90	0.00	0.00	0.00	n/a
server_recount	under_report_input_50	0.00	0.00	0.00	n/a
server_recount	drop_reasoning	0.00	0.00	0.00	n/a
server_recount	inflate_cached	0.00	0.00	0.00	n/a
server_recount	total_mismatch	0.00	0.00	0.00	n/a
server_recount	rounding_shave	0.00	0.00	0.00	n/a
hybrid_reconcile	honest	0.00	0.00	0.00	n/a
hybrid_reconcile	under_report_output_50	0.00	0.00	0.00	n/a
hybrid_reconcile	under_report_output_90	0.00	0.00	0.00	n/a
hybrid_reconcile	under_report_input_50	0.00	0.00	0.00	n/a
hybrid_reconcile	drop_reasoning	0.00	0.00	0.00	n/a
hybrid_reconcile	inflate_cached	0.00	0.00	0.00	n/a

Table 8: M2: leakage efficiency vs concurrency (attacker arm; volume held constant).

architecture	c=1	c=5	c=20	c=50	c=100
client	0.228	0.228	0.228	0.228	0.228
client_logged	0.228	0.228	0.228	0.228	0.228
client_total	0.257	0.257	0.257	0.257	0.257
hybrid_reconcile	0.000	0.000	0.000	0.000	0.000
server_recount	0.000	0.000	0.000	0.000	0.000
upstream	0.000	0.000	0.000	0.000	0.000

Table 9: M2 as a transformation on the usage vector. Whether a manipulation pays off is determined by whether the pricing basis reads the field T changes.

architecture	pricing basis	manipulation T	vector change T(U)	leak eff.	why it succeeds / fails
client	category	honest	unchanged	+0.000	T is identity (honest)
client	category	under_report_output_50	output:60->30, total:98->68	+0.324	T lowers a category the basis prices
client	category	under_report_output_90	output:60->6, total:98->44	+0.583	T lowers a category the basis prices
client	category	under_report_input_50	input:8->4, total:98->94	+0.014	T lowers a category the basis prices
client	category	drop_reasoning	reasoning:30->0, total:98->68	+0.324	T lowers a category the basis prices
client	category	inflate_cached	output:60->30, cached_input:0->30	+0.313	T lowers a category the basis prices
client	category	total_mismatch	total:98->24	+0.000	T changes only the declared total; category basis never reads it
client	category	rounding_shave	input:8->7, output:60->57, total:98->94	+0.036	T lowers a category the basis prices
client_total	flat-total	honest	unchanged	-0.058	T is identity (honest)
client_total	flat-total	under_report_output_50	output:60->30, total:98->68	+0.266	T lowers the declared total the basis prices
client_total	flat-total	under_report_output_90	output:60->6, total:98->44	+0.525	T lowers the declared total the basis prices
client_total	flat-total	under_report_input_50	input:8->4, total:98->94	-0.014	T's change is priced at (near) the same rate
client_total	flat-total	drop_reasoning	reasoning:30->0, total:98->68	+0.266	T lowers the declared total the basis prices
client_total	flat-total	inflate_cached	output:60->30, cached_input:0->30	-0.058	T changes only subtotals; flat-total basis never reads them
client_total	flat-total	total_mismatch	total:98->24	+0.741	T lowers the declared total the basis prices
client_total	flat-total	rounding_shave	input:8->7, output:60->57, total:98->94	-0.014	T's change is priced at (near) the same rate

Table 10: Economic sensitivity. Dollar columns are SCENARIO values under synthetic price tiers; leakage efficiency and leaked tokens are the price-independent measures. Efficiency is invariant under uniform price scaling (low->medium) but not under a price-structure change (medium->high alters the output/input ratio).

architecture	manipulation	low (\$)	medium (\$)	high (\$)	leaked tokens	leak eff.	inv. under x10 scale?	d(ef) med->high
client	honest	+0.0000	+0.0000	+0.0000	0	+0.000	yes	+0.000
client_total	honest	-0.0160	-0.1600	-1.9200	-113	-0.058	yes	-0.012
client	under_report_output_50	+0.0900	+0.9000	+9.0000	635	+0.324	yes	+0.004
client_total	under_report_output_50	+0.0740	+0.7400	+7.0800	522	+0.266	yes	-0.009
client	under_report_output_90	+0.1620	+1.6200	+16.2000	1142	+0.583	yes	+0.007
client_total	under_report_output_90	+0.1460	+1.4600	+14.2800	1029	+0.525	yes	-0.006
client	under_report_input_50	+0.0040	+0.0400	+0.2400	28	+0.014	yes	-0.006
client_total	under_report_input_50	-0.0040	-0.0400	-0.7200	-28	-0.014	yes	-0.012
client	drop_reasoning	+0.0900	+0.9000	+9.0000	635	+0.324	yes	+0.004
client_total	drop_reasoning	+0.0740	+0.7400	+7.0800	522	+0.266	yes	-0.009
client	inflate_cached	+0.0870	+0.8700	+8.8200	613	+0.313	yes	+0.008
client_total	inflate_cached	-0.0160	-0.1600	-1.9200	-113	-0.058	yes	-0.012
client	total_mismatch	+0.0000	+0.0000	+0.0000	0	+0.000	yes	+0.000
client_total	total_mismatch	+0.2060	+2.0600	+20.2800	1452	+0.741	yes	-0.003
client	rounding_shave	+0.0100	+0.1000	+0.9600	71	+0.036	yes	-0.001
client_total	rounding_shave	-0.0040	-0.0400	-0.7200	-28	-0.014	yes	-0.012

Table 11: M1 ablation: abort-inclusive finalization closes the integrity violation; reservation independently provides the solvency (funds) guarantee. Both are needed for a correct design.

architecture	reservation	abort-finalization	worst leak/req	integrity	served (budget 2)	solvency
pre_debit	yes	n/a (pre-committed)	-0.0090	held	2/6	ok
reserve_reconcile	yes	yes	+0.0000	held	2/6	ok
reserve_refund_on_abort	yes	no	+0.0845	VIOLATED	6/6	ok
no_reserve_settle	no	yes	+0.0000	held	6/6	NEGATIVE
post_completion	no	no	+0.0845	VIOLATED	6/6	ok

Table 12: M2 ablation: performing a recount is not sufficient; it must be used as the billing basis. client_logged computes the recount and still leaks.

architecture	recount performed	recount used for billing	worst leak eff.	integrity	detection
client	no	no	0.595	VIOLATED	D0
client_logged	yes	no	0.595	VIOLATED	D1
client_total	no	no	0.745	VIOLATED	D0
server_recount	yes	yes	0.000	held	n/a
hybrid_reconcile	yes	yes	0.000	held	n/a
upstream	no	no	0.000	held	n/a

Table 13: Lifecycle failure injection. Every invariant violation occurs on a vulnerable architecture; the hardened architectures survive all injected failures, and a recount-engine failure is fail-closed.

id	failure injected	architecture	delivered	net debit	refund	leakage	invariant
F1	client disconnect mid-stream	post_completion	yes	0.0000	0.0000	0.2375	VIOLATED
F1	client disconnect mid-stream	reserve_reconcile	yes	0.2375	0.2175	0.0000	HELD
F2	client disconnect before first token	post_completion	yes	0.0000	0.0000	0.0275	VIOLATED
F2	client disconnect before first token	reserve_reconcile	yes	0.0275	0.4275	0.0000	HELD
F8	partial/truncated stream near completion	post_completion	yes	0.0000	0.0000	0.4325	VIOLATED
F8	partial/truncated stream near completion	reserve_reconcile	yes	0.4325	0.0225	0.0000	HELD
F9	audit read before settle grace period	post_completion	yes	0.0000	0.0000	0.2375	VIOLATED
F9	audit read before settle grace period	reserve_reconcile	yes	0.2375	0.2175	0.0000	HELD
F5	missing/unknown usage metadata	client	yes	0.6725	0.0000	0.0000	HELD
F5	missing/unknown usage metadata	server_recount	yes	0.6725	0.0000	0.0000	HELD
F7	duplicate usage metadata (replayed request_i	client	yes	0.2750	0.0000	0.3975	VIOLATED
F7	duplicate usage metadata (replayed request_i	server_recount	yes	0.6725	0.0000	0.0000	HELD
F6	delayed/out-of-order usage metadata	client	yes	0.4550	0.0000	0.2175	VIOLATED
F6	delayed/out-of-order usage metadata	server_recount	yes	0.6725	0.0000	0.0000	HELD
F4	tokenizer/recount engine unavailable	bench/server_recount	no	0.0000	0.0000	0.0000	HELD

Table 14: Detectability matrix. D0 invisible; D1 only via post-hoc reconciliation; D2 logged discrepancy; D3 prevented or corrected in real time. *B0's final balance alone cannot reveal the loss because the ledger and the balance are both under-decremented consistently with the surviving write.

mech	architecture	req logs	usage ledger	reconciliation	final balance	real-time	level	detection latency
B0	vulnerable (non-atomic)	no	yes	yes	no*	no	D1	until a reconciliation pass
B0	hardened (atomic CAS)	n/a	n/a	n/a	n/a	yes	D3	0 (prevented)
M1	post_completion	partial	no	no	no	no	D0	never (no record is written)
M1	reserve_refund_on_abort	partial	yes	yes	yes	no	D1	until a reconciliation pass
M1	reserve_reconcile	n/a	n/a	n/a	n/a	yes	D3	0 (prevented)
M2	client	no	no	no	no	no	D0	never (ledger records the lie)
M2	client_logged	no	yes	yes	no	no	D1	until a reconciliation pass
M2	server_recount	n/a	n/a	n/a	n/a	yes	D3	0 (prevented)
M2	hybrid_reconcile	yes	yes	yes	n/a	yes	D3	0 (detected and corrected in-request)

Table 15: Architecture safety summary across all attacks and tiers.

mech	architecture	verdict	max req ASR	best detect
M1	pre_debit	safe	0.00	D3
M1	reserve_reconcile	safe	0.00	D3
M1	post_completion	LEAKS	1.00	D0
M1	reserve_refund_on_abort	LEAKS	1.00	D0
M2	client	LEAKS	1.00	D0
M2	client_logged	LEAKS	1.00	D1
M2	client_total	LEAKS	1.00	D0
M2	upstream	safe	0.00	D3
M2	server_recount	safe	0.00	D3
M2	hybrid_reconcile	safe	0.00	D3

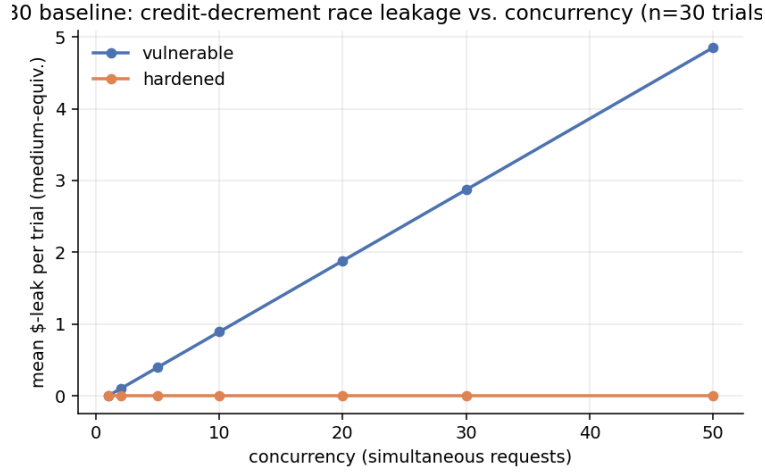


Figure 5: B0 baseline: mean \$-leak per trial vs. concurrency (30 repetitions per cell). Leakage is *created* by concurrency, in contrast with M1/M2.

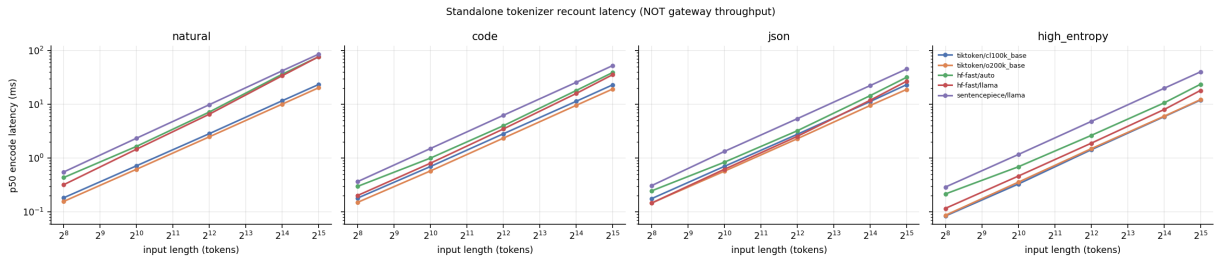


Figure 6: Standalone tokenizer recount latency (p50, log-log) by engine, workload and input length, over 100 exact-length-verified cells. This is *not* a gateway throughput result.